HUGO MOSSER

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Open to relocation worldwide

SUMMARY

AI/ML Engineer with a French Engineering degree (Master of Engineering) in Computer Science with AI specialization and a track record of shipping AI systems end-to-end: from research prototypes to deployed APIs. Skilled in Python, PyTorch, TensorFlow, FastAPI and cloud-based model serving (GCP, TF Serving), with experience spanning NLP, Computer Vision and LLM integration (LangChain, RAG, OpenAI API). Built and published multiple production-oriented projects independently. International research background (Osaka Metropolitan University, Japan). Open to relocation worldwide.

WORK EXPERIENCE

AI Engineer – Independent Projects Sep 2025 – Present
Self-directed · Remote

- Built an end-to-end plant disease classification system using a CNN (TensorFlow/Keras) trained on 3-class image data (Early Blight, Late Blight, Healthy); served via TF Serving + FastAPI backend and a React Native mobile app with real-time camera inference; infrastructure configured on GCP - [reaching 96% test accuracy]
- Designed a hybrid log classification pipeline combining Regex rules, Sentence Transformers + Logistic Regression, and LLM fallback to handle logs of varying complexity; deployed as a FastAPI microservice
- Developed a RAG-based news research tool using LangChain, OpenAI API and Streamlit, enabling semantic search and insight extraction over live news sources
- Delivered all projects with public GitHub repositories, documented READMEs, and deployable backends demonstrating full ML lifecycle from data to production : https://github.com/strandedDepths

AI R&D Engineer Intern Apr 2024 – Oct 2024
STMicroelectronics · Crolles · France

- Developed neural surrogate models in PyTorch to approximate computationally expensive analog circuit simulations, enabling faster design iteration cycles
- Built end-to-end data preprocessing and training pipelines in Python, handling domain specific engineering data from raw inputs to model-ready tensors
- Evaluated model performance against real-world design cases in collaboration with an engineering team, iterating on architecture and training strategy to meet accuracy requirements

ML/AI Research Scientist Intern Apr 2023 – Sep 2023
Osaka Metropolitan University · Osaka, Japan

- Conducted research on multimodal AI for onomatopoeia generation: designed a pipeline mapping visual inputs (images) to textual sound representations using deep learning models in TensorFlow
- Implemented and benchmarked image captioning and sequence generation architectures, adapting them to a low-resource cross-lingual setting (Japanese/English)
- Conducted full experimental lifecycle: dataset curation, model training, evaluation, and results analysis in an international academic research environment

TECHNICAL SKILLS

Languages	Python, C, C++, Java
ML / Deep Learning	TensorFlow/Keras, PyTorch, scikit-learn, Hugging Face Transformers, Pandas, NumPy
LLM / GenAI	LangChain, OpenAI API, RAG pipelines, Prompt Engineering, Dialogflow
Backend / Serving	FastAPI, TF Serving, MySQL, React.js, React Native, MongoDB, Streamlit
Infrastructure / MLOps	GCP, Docker, GCP, CI/CD, Linux, Git
Foundation	Algorithms, Optimization, Statistics, Signal Processing

EDUCATION

Master's-level Engineering Degree in Computer Science Sep 2019 – Sep 2024
CY Tech, CY Cergy Paris Université · France

Specialization in Artificial Intelligence · French Grande École (Master of Science in Computer Science Engineering)

Master's Thesis: Automatic Music Transcription - Designed an end-to-end system combining transformer architectures and reservoir computing to transcribe audio signals into musical scores

Baccalauréat S (A-levels equivalent) July 2019
Lycée L'Immaculée Conception · France

ADDITIONAL INFORMATION

Languages: French (native), English (C1 · TOEIC 960), Japanese (B1), Spanish (B1), German (A1), Chinese(A1)
Leadership: President of 3 student organizations during engineering studies (Hiking, Japanese Culture, Arts) managed events and member communities of 10–20 people
Teaching: Volunteer Japanese language instructor for engineering students · 2 years
Interests: Trail, running, swimming, drawing, piano